Seneca

SEG800
Digital Signal and Image Processing

Image Processing in Frequency Domain

Seneca College

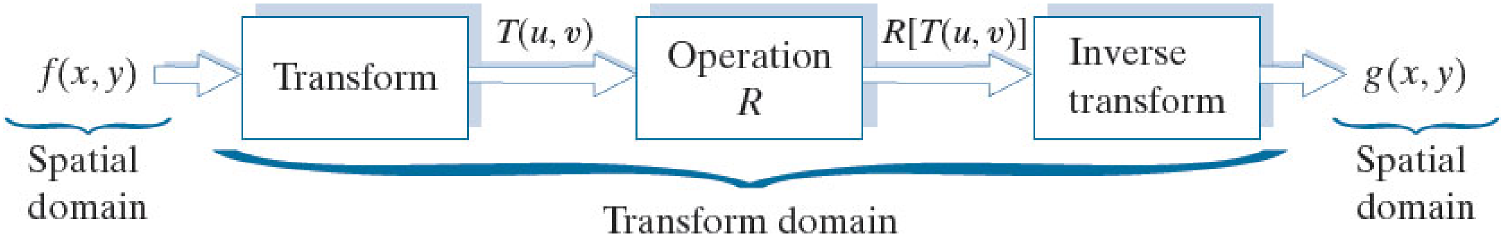
[Slides adapted from references on the last slide]

Overview

- Why image processing in frequency domain
- 2-D sampling theorem
- 2-D DFT and IDFT
- 2-D Discrete Convolution
- Filtering in the frequency domain

Reminder: Figure 2.44

General approach for working in the linear transform domain.



Feasibility of Image Processing in Frequency Domain

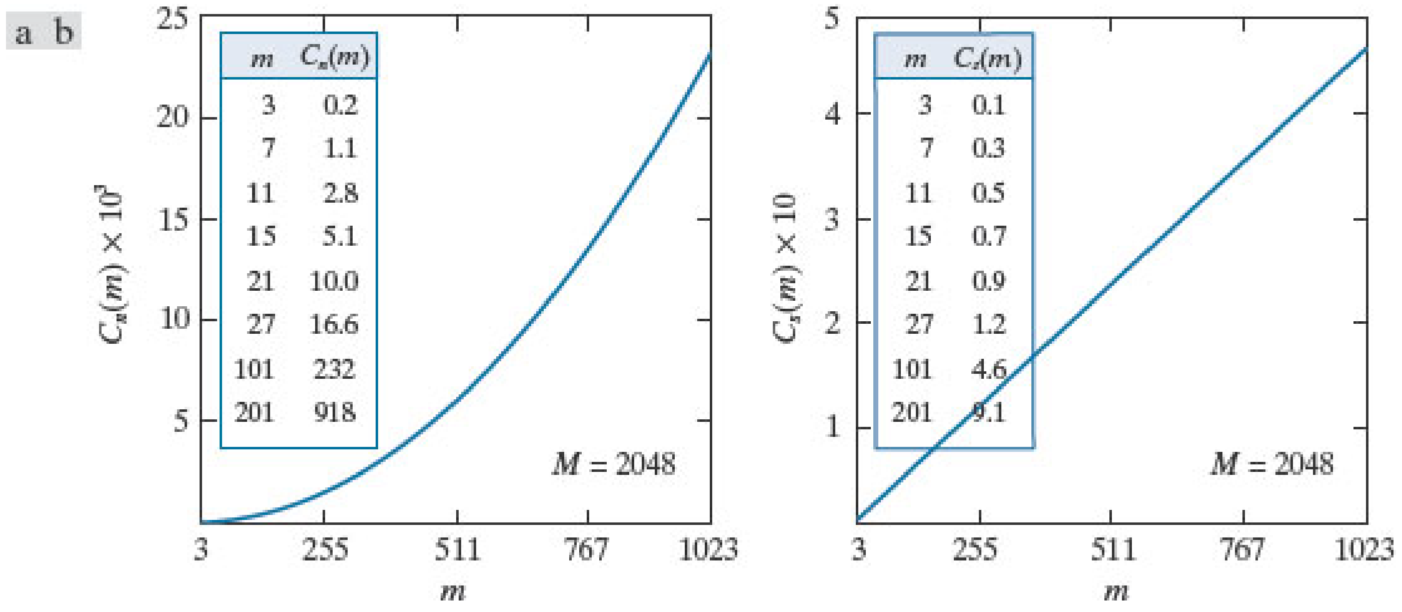
- Filtering an M×N image by a m×n kernel

Filtering	Order of operations
Spatial filtering	MNmn
Spatial filtering with separable kernel	MN(m+n)
Filtering in frequency domain using FFT	$2MN\log_2 MN$

- Computational advantage (assuming M & m)
  - over spatial filtering: $$C_n(m) = \frac{M^2m^2}{2M^2\log_2 M^2} = \frac{m^2}{4\log_2 M}$$
  - over separable spatial filtering: $$C_s(m) = \frac{2M^2m}{2M^2\log_2 M^2} = \frac{m}{2\log_2 M}$$

Figure 4.2

(a) Computational advantage of the FFT over nonseparable spatial kernels. (b) Advantage over separable kernels. The numbers for C(m) in the inset tables are not to be multiplied by the factors of 10 shown for the curves.



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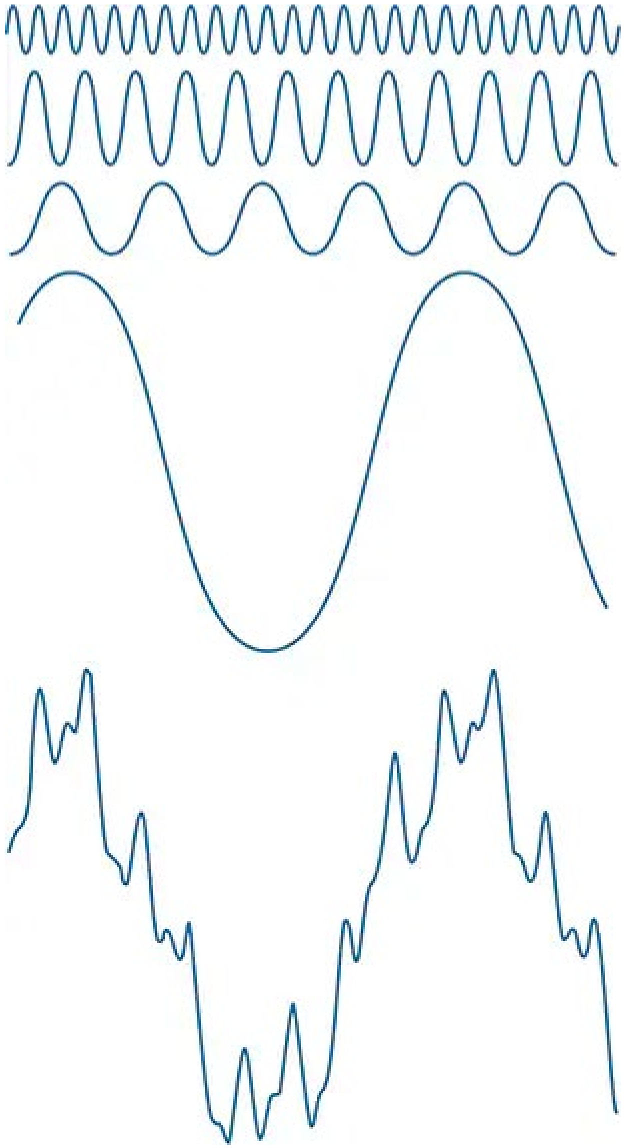
Reminder: Fourier Transform

- **Fourier series**: Any periodic function can be expressed as the sum of sines and/or cosines of different frequencies, each multiplied by a different coefficient.

- **Fourier transform**: Functions that are not periodic (but whose area under the curve is finite) can be expressed as the integral of sines and/or cosines multiplied by a weighting function.



Reminder- Convolution

- The Fourier transform of the convolution of two functions in the spatial domain is equal to the product in the frequency domain of the Fourier transforms of the two functions.
- Conversely, if we have the product of the two transforms, we can obtain the convolution in the spatial domain by computing the inverse Fourier transform.
- In other words, $f \star h$ and $H \cdot F$ are a Fourier transform pair.

$$(f \bigstar h)(t) \Leftrightarrow (H \cdot F)(\mu)$$

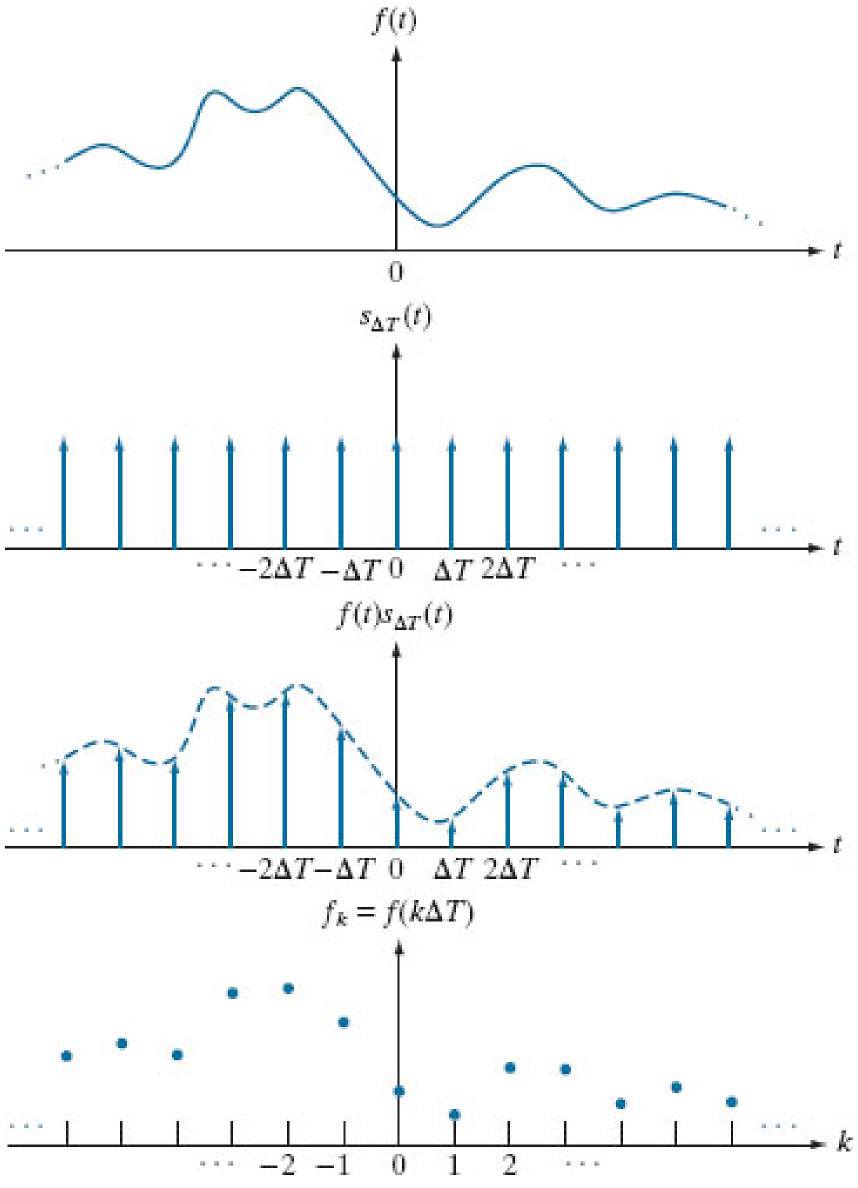
- Also, convolution in the frequency domain is analogous to multiplication in the spatial domain

$$(f \cdot h)(t) \Leftrightarrow (H \bigstar F)(\mu)$$

Figure 4.5

a
b
c
d

(a) A continuous function. (b) Train of impulses used to model sampling. (c) Sampled function formed as the product of (a) and (b). (d) Sample values obtained by integration and using the sifting property of impulses. (The dashed line in (c) is shown for reference. It is not part of the data.)



Unit impulse, delta function, or Dirac delta function:

$$\delta(t) = \begin{cases} \infty & \text{if } t = 0 \\ 0 & \text{if } t \neq 0 \end{cases} \qquad \int_{-\infty}^{\infty} \delta(t)dt = 1$$

Sifting Property:

$$\int_{-\infty}^{\infty} f(t)\delta(t)dt = f(0)$$

Impulse Train:

$$s_{\Delta T}(t) = \sum_{k=-\infty}^{\infty} \delta(t - k\Delta T)$$

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Reminder- Sampling

- To be processed in a digital computer, continuous functions need to be sampled into a sequence of discrete values.

$$\tilde{f}(t) = f(t)s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} f(t)\delta(t - n\Delta T)$$

- When working with discrete variables:

$$\delta(x) = \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{if } x \neq 0 \end{cases} \qquad \sum_{x=-\infty}^{\infty} \delta(x) = 1$$

- Sifting property:

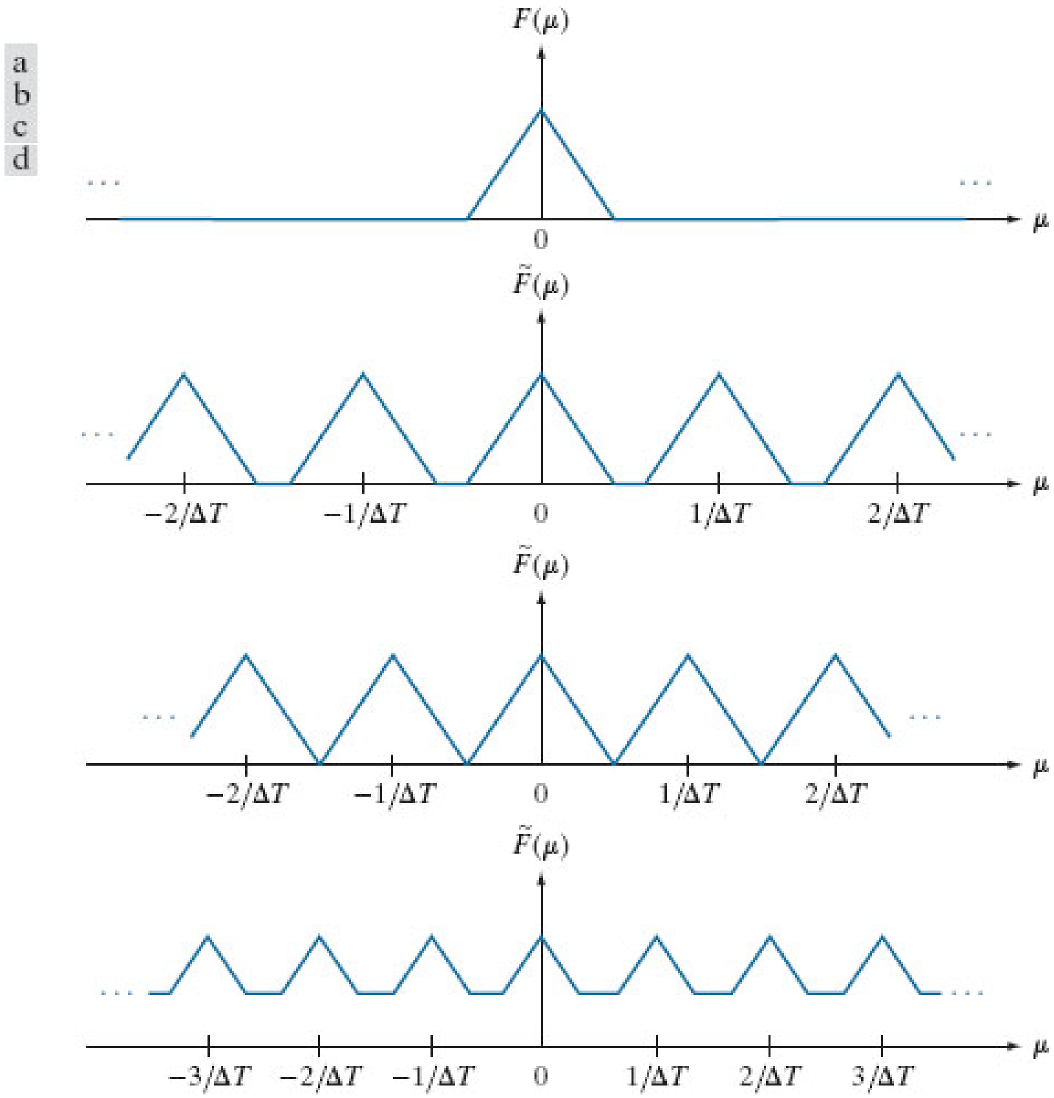
$$\sum_{x=-\infty}^{\infty} f(x)\delta(x) = f(0)$$

Fourier Transform of Sampled Functions

The Fourier transform $\tilde{F}(\mu)$ of the sampled function $\tilde{f}(t)$ is an *infinite, periodic* sequence of copies of the transform of the original, continuous function. The separation between copies is determined by the value of $1/\Delta T$.

Observe that although $\tilde{f}(t)$ is a sampled function, its transform, $\tilde{F}(\mu)$, is continuous because it consists of copies of $F(\mu)$, which is a continuous function.

Figure 4.6. (a) Illustrative sketch of the Fourier transform of a band-limited function. (b)–(d) Transforms of the corresponding sampled functions under the conditions of over-sampling, critically sampling, and under-sampling, respectively.



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Reminder: Sampling Theorem

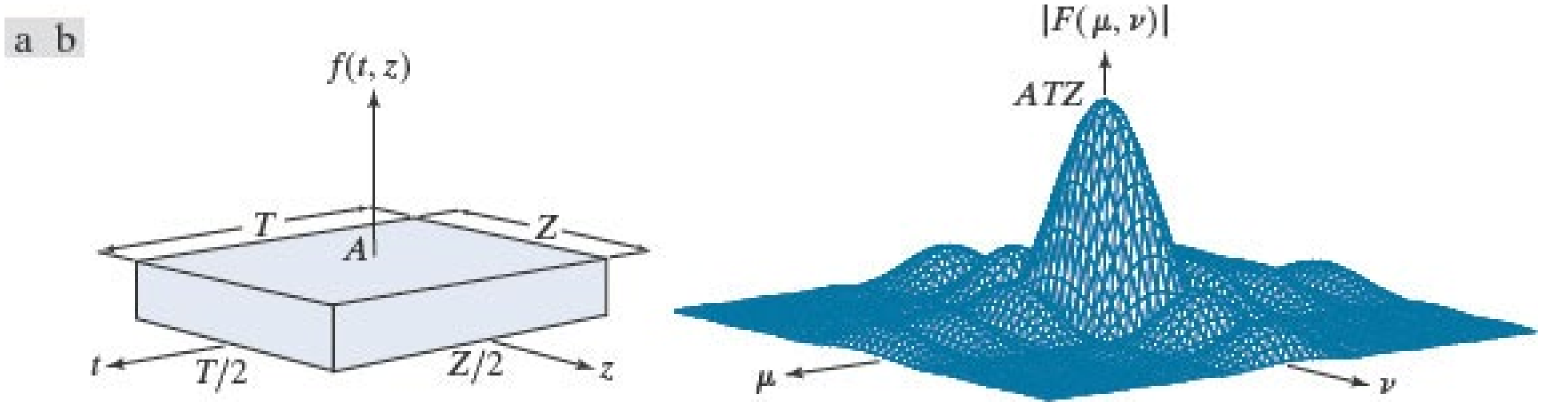
- A continuous, *band-limited* function can be recovered completely from a set of its samples if the samples are acquired at a rate exceeding twice the highest frequency content of the function.

$$\frac{1}{\Delta T} > 2\mu_{\max}$$

- A sampling rate exactly equal to twice the highest frequency is called the Nyquist rate.

Figure 4.14

(a) A 2-D function and (b) a section of its spectrum. The box is longer along the **t**-axis, so the spectrum is more contracted along the μ -axis



2-D Sampling and 2-D Sampling Theorem

- Multiplying f(t,z) by a 2D impulse train yields the sampled function.

$$s_{\Delta T \Delta Z}(t,z) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \delta(t - m\Delta T, z - n\Delta Z)$$

- Function f(t,z) is band-limited if its Fourier transform is 0 outside a limited interval:

$$F(\mu,\nu) = 0 \qquad \text{for } |\mu| \geq \mu_{\max} \text{ and } |\nu| \geq \nu_{\max}$$

- 2-D Sampling Theorem: A continuous band-limited function f(t,z) can be recovered with no error from its samples, if the sampling rates/intervals follow these conditions:

$$F_T = \frac{1}{\Delta T} > 2\mu_{max} \quad \text{and} \quad F_Z = \frac{1}{\Delta Z} > 2\nu_{max}$$

Figure 4.15

2-D impulse train.

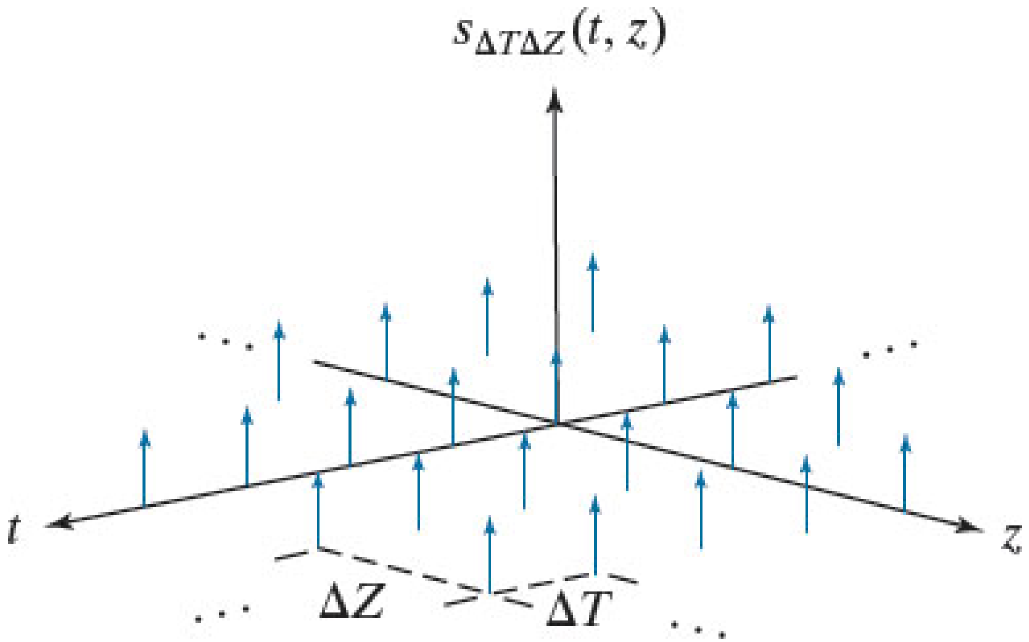


Figure 4.16

Two-dimensional Fourier transforms of (a) an over-sampled, and (b) an under-sampled, band-limited function.

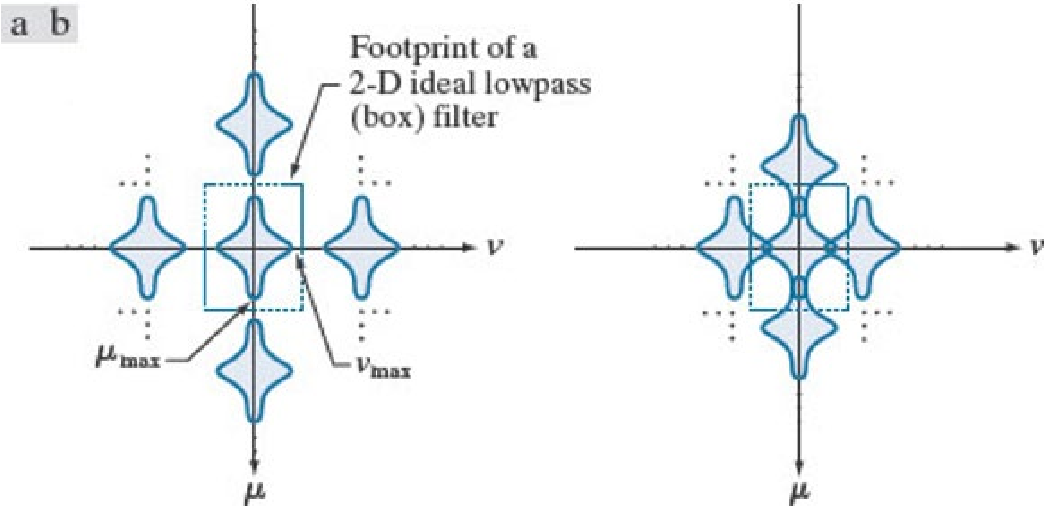


Figure 4.17

Various aliasing effects resulting from the interaction between the frequency of 2-D signals and the sampling rate used to digitize them. The regions outside the sampling grid are continuous and free of aliasing.

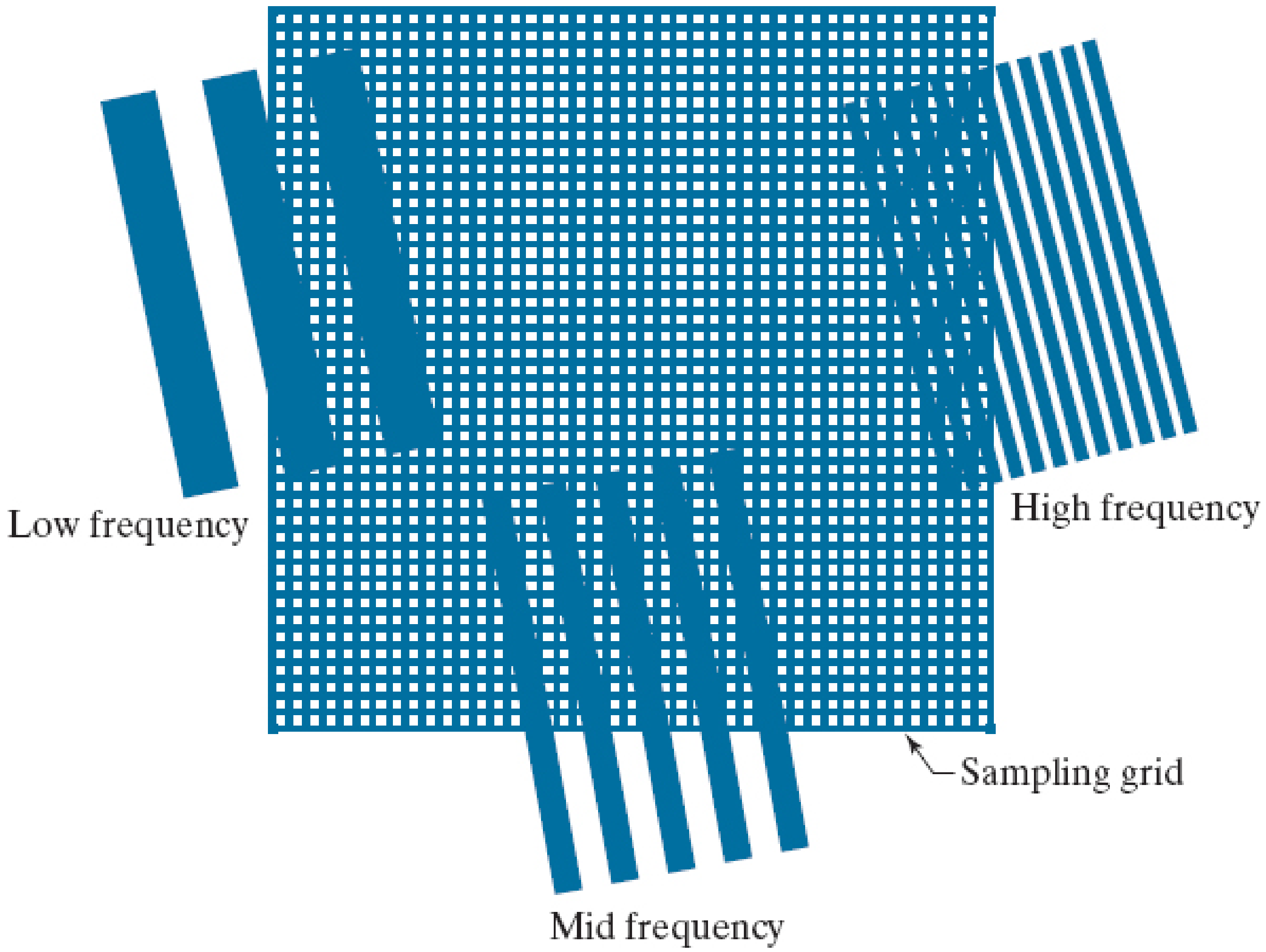


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Figure 4.18

Aliasing. In (a) and (b) the squares are of sizes 16 and 6 pixels on the side. In (c) and (d) the squares are of sizes 0.95 and 0.48 pixels, respectively. Each small square in (c) is one pixel. Both (c) and (d) are aliased. Note how (d) masquerades as a “normal” image.

a b
c d

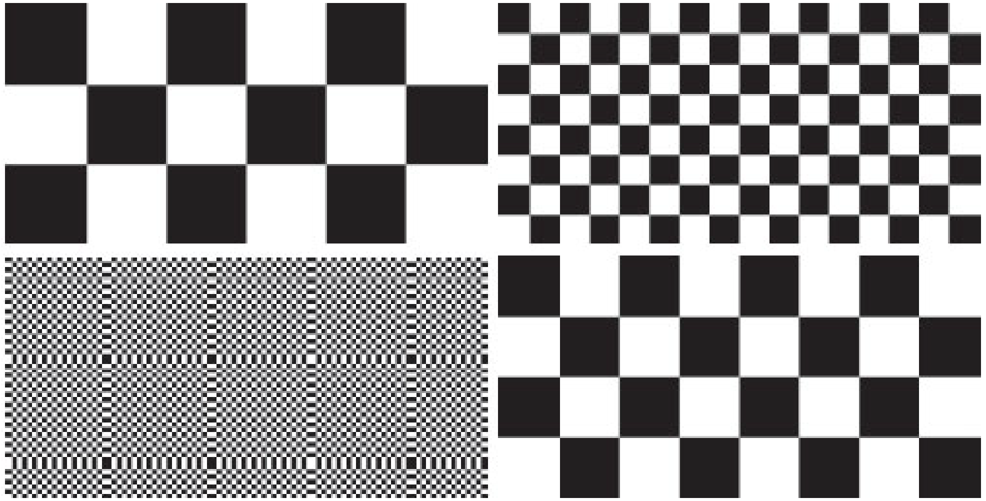
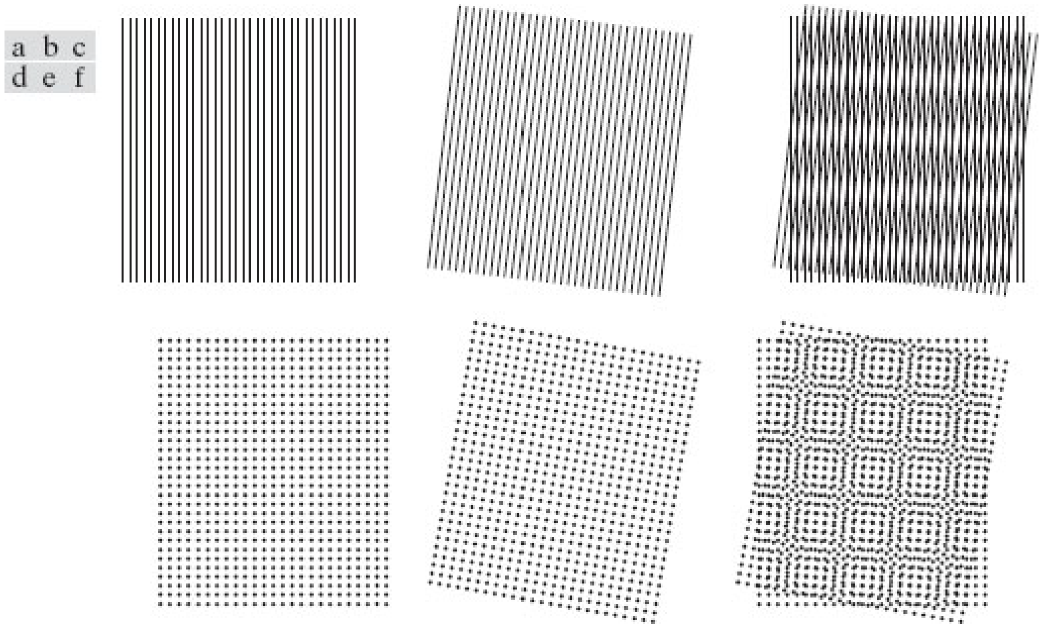


Figure 4.20

Examples of the moiré effect. These are vector drawings, not digitized patterns. Superimposing one pattern on the other is analogous to multiplying the patterns.



2-D Discrete Fourier Transform

- One variable (M samples)

$$F(u) = \sum_{x=0}^{M-1} f(x) e^{-j2\pi ux/M}$$

$$u = 0, 1, 2, \ldots, M-1$$

$$f(x) = \frac{1}{M} \sum_{u=0}^{M-1} F(u) e^{j2\pi ux/M}$$

$$x = 0, 1, 2, \ldots, M-1$$

- Two variables (MxN samples)

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v)\, e^{j2\pi(ux/M + vy/N)}$$

Properties of 2-D DFT & IDFT

Spatial vs. Frequency Intervals

- Suppose that a continuous function f(t, z) is sampled to form a digital image, f(x, y), consisting of samples taken in the t- and z-directions, respectively. Let ΔT and ΔZ denote the separations between samples.
- The separations between samples in the frequency domain are inversely proportional both to the spacing between spatial samples and to the number of samples.

$$\Delta u = \frac{1}{M\Delta T} \qquad \Delta v = \frac{1}{N\Delta Z}$$

Translation & Rotation

$$f(x,y)e^{j2\pi(u_0x/M+v_0y/N)} \Leftrightarrow F(u-u_0, v-v_0)$$

$$f(x-x_0, y-y_0) \Leftrightarrow F(u,v)e^{-j2\pi(x_0u/M+y_0v/N)}$$

- Using polar coordinates

$$x = r\cos\theta \qquad y = r\sin\theta \qquad u = \omega\cos\varphi \qquad v = \omega\sin\varphi$$

$$f(r, \theta+\theta_0) \Leftrightarrow F(\omega, \varphi+\theta_0)$$

- Thus, rotating f(x,y) by an angle θ_0 rotates F(u,v) by the same angle, and vice versa.

Periodicity & Centering

- The 2-D Fourier transform and its inverse are **infinitely periodic** in the u and v directions (as in 1-D):

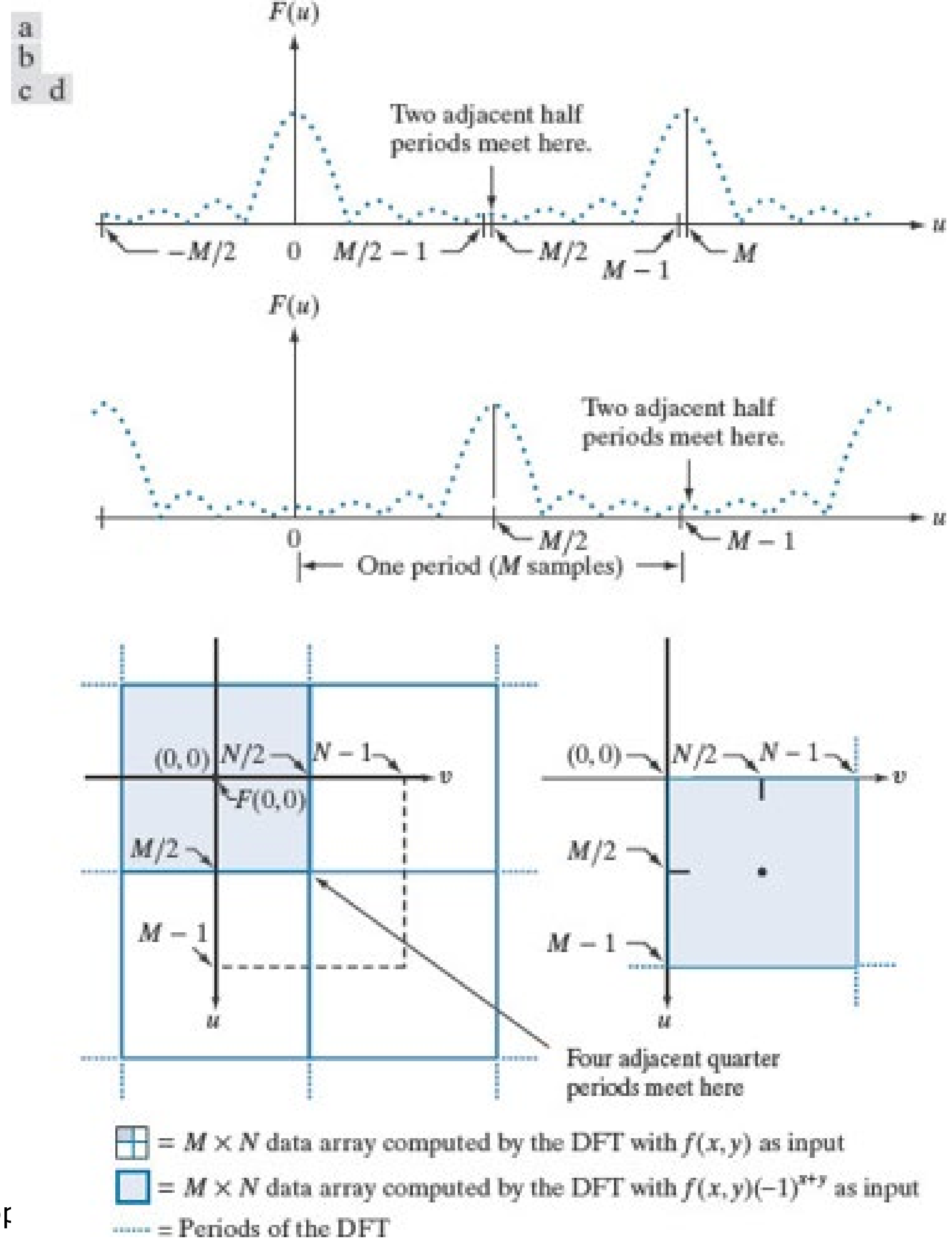
$$F(u,v) = F(u + k_1 M, v) = F(u, v + k_2 N) = F(u + k_1 M, v + k_2 N)$$

$$f(x,y) = f(x + k_1 M, y) = f(x, y + k_2 N) = f(x + k_1 M, y + k_2 N)$$

- Centering:
  - Since $f(x)e^{j2\pi(u_0 x/M)} \Leftrightarrow F(u - u_0)$, we have $f(x)(-1)^x \Leftrightarrow F(u - M/2)$
  - Multiplying f(x) by $(-1)^x$ shifts the data, centering F(u) on the interval [0, M-1]

Figure 4.22

Centering the Fourier transform. (a) A 1-D DFT showing an infinite number of periods. (b) Shifted DFT obtained by multiplying $f(x)$ by -1^x before computing $F(u)$. (c) A 2-D DFT showing an infinite number of periods. The area within the dashed rectangle is the data array, $F(u,v)$, obtained with Eq. (4-67) with an image $f(x,y)$ as the input. This array consists of four quarter periods. (d) Shifted array obtained by multiplying $f(x,y)$ by -1^{x+y} before computing $F(u,v)$. The data now contains one complete, centered period, as in (b).



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Symmetry

- For even and odd components of function w:

$$w(x,y) = w_e(x,y) + w_o(x,y)$$

$$w_e(x,y) \triangleq \frac{w(x,y) + w(-x,-y)}{2} \qquad w_o(x,y) \triangleq \frac{w(x,y) - w(-x,-y)}{2}$$

- Even components are symmetric; odd components antisymmetric:

$$w_e(x,y) = w_e(-x,-y) = w_e(M-x, N-y)$$
$$w_o(x,y) = -w_o(-x,-y) = -w_o(M-x, N-y)$$

- The Fourier transform of a real function is conjugate symmetric:

$$F^*(u,v) = F(-u,-v)$$

Table 4.1

Some symmetry properties of the 2-D DFT and its inverse. **R(u,v)** and **I(u,v)** are the real and imaginary parts of **F(u,v),** respectively. Use of the word complex indicates that a function has nonzero real and imaginary parts.

	Spatial Domain†		Frequency Domain†
1)	$f(x,y)$ real	$\Leftrightarrow$	$F^*(u,v) = F(-u,-v)$
2)	$f(x,y)$ imaginary	$\Leftrightarrow$	$F^*(-u,-v) = -F(u,v)$
3)	$f(x,y)$ real	$\Leftrightarrow$	$R(u,v)$ even; $I(u,v)$ odd
4)	$f(x,y)$ imaginary	$\Leftrightarrow$	$R(u,v)$ odd; $I(u,v)$ even
5)	$f(-x,-y)$ real	$\Leftrightarrow$	$F^*(u,v)$ complex
6)	$f(-x,-y)$ complex	$\Leftrightarrow$	$F(-u,-v)$ complex
7)	$f^*(x,y)$ complex	$\Leftrightarrow$	$F^*(-u,-v)$ complex
8)	$f(x,y)$ real and even	$\Leftrightarrow$	$F(u,v)$ real and even
9)	$f(x,y)$ real and odd	$\Leftrightarrow$	$F(u,v)$ imaginary and odd
10)	$f(x,y)$ imaginary and even	$\Leftrightarrow$	$F(u,v)$ imaginary and even
11)	$f(x,y)$ imaginary and odd	$\Leftrightarrow$	$F(u,v)$ real and odd
12)	$f(x,y)$ complex and even	$\Leftrightarrow$	$F(u,v)$ complex and even
13)	$f(x,y)$ complex and odd	$\Leftrightarrow$	$F(u,v)$ complex and odd

Fourier Spectrum and Phase Angle

- In Polar Form: $F(u,v) = R(u,v) + jI(u,v) \; = |F(u,v)|\, e^{j\phi(u,v)}$
- Magnitude: $|F(u,v)| = \left[R^2(u,v) + I^2(u,v)\right]^{1/2}$
- Phase angle/spectrum: $\phi(u,v) = \arctan\left[\dfrac{I(u,v)}{R(u,v)}\right]$
- Power Spectrum: $P(u,v) = |F(u,v)|^2$
- DC Component: $F(0,0) = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y) \; = MN\bar{f}$

Figure 4.23

(a) Image. (b) Spectrum, showing small, bright areas in the four corners (you have to look carefully to see them). (c) Centered spectrum. (d) Result after a log transformation. The zero crossings of the spectrum are closer in the vertical direction because the rectangle in (a) is longer in that direction. The right-handed coordinate convention used in the book places the origin of the spatial and frequency domains at the top left (see Fig. 2.19).

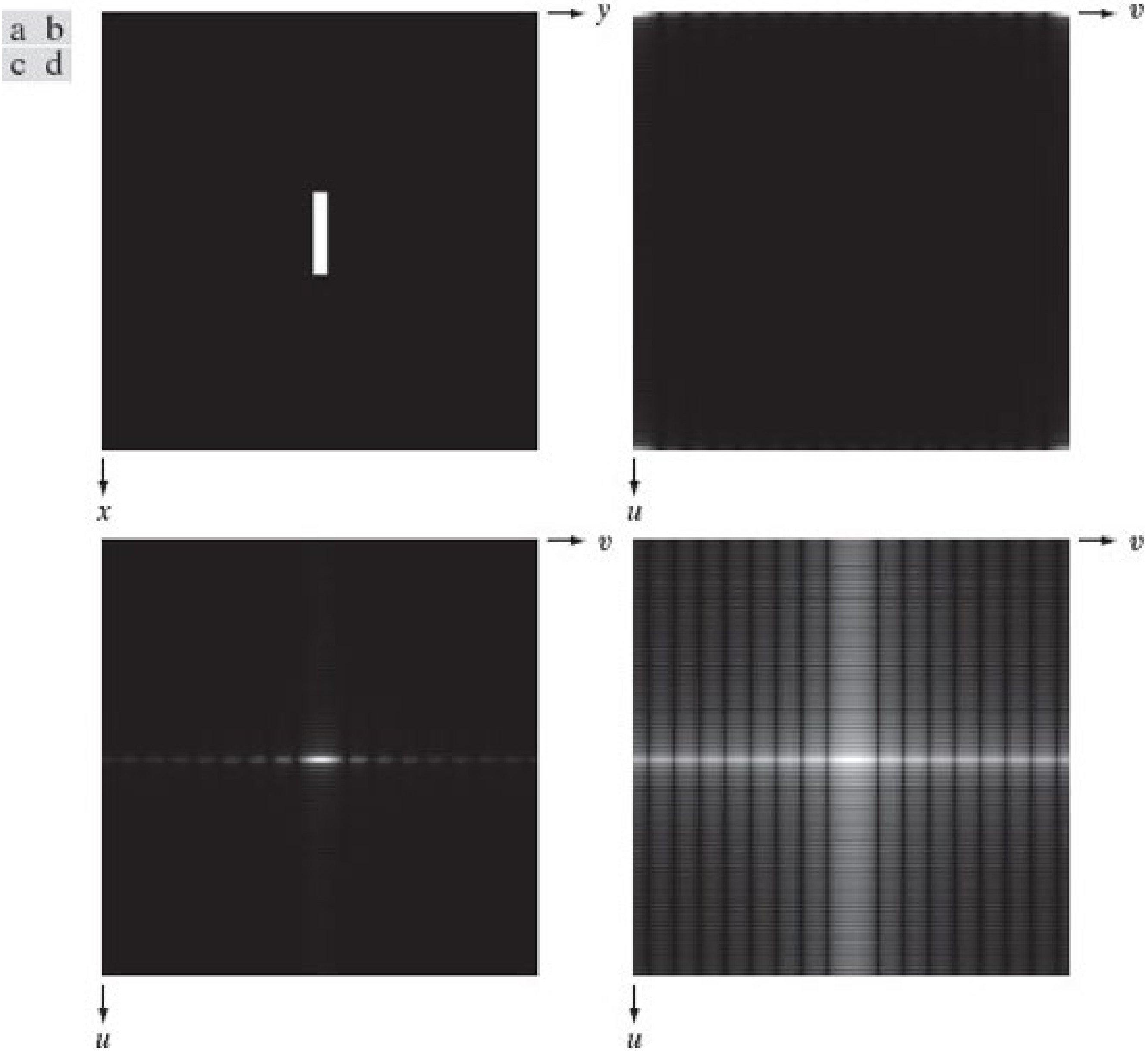

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Figure 4.24

(a) The rectangle in Fig. 4.23(a) translated. (b) Corresponding spectrum. (c) Rotated rectangle. (d) Corresponding spectrum. The spectrum of the translated rectangle is identical to the spectrum of the original image in Fig. 4.23(a).

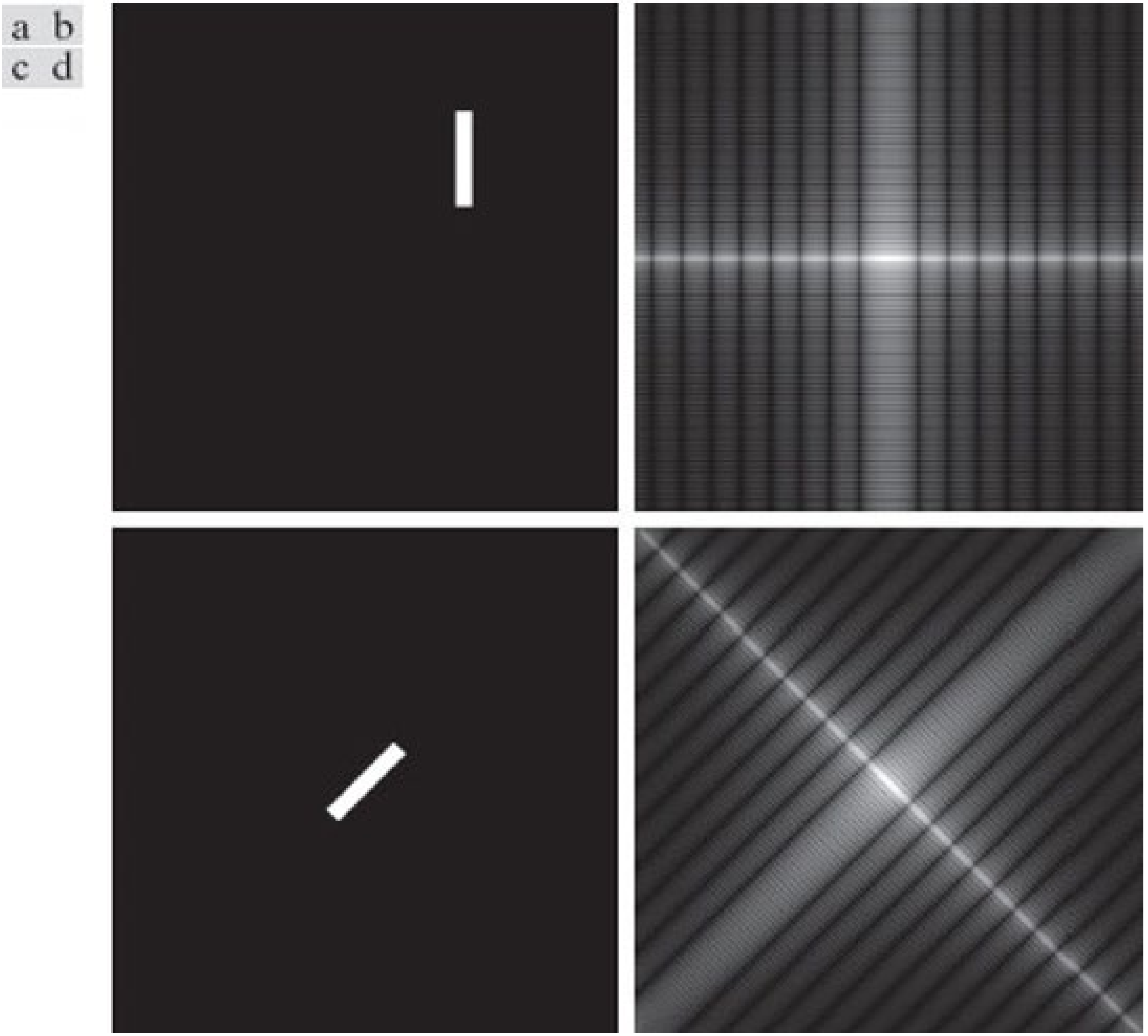
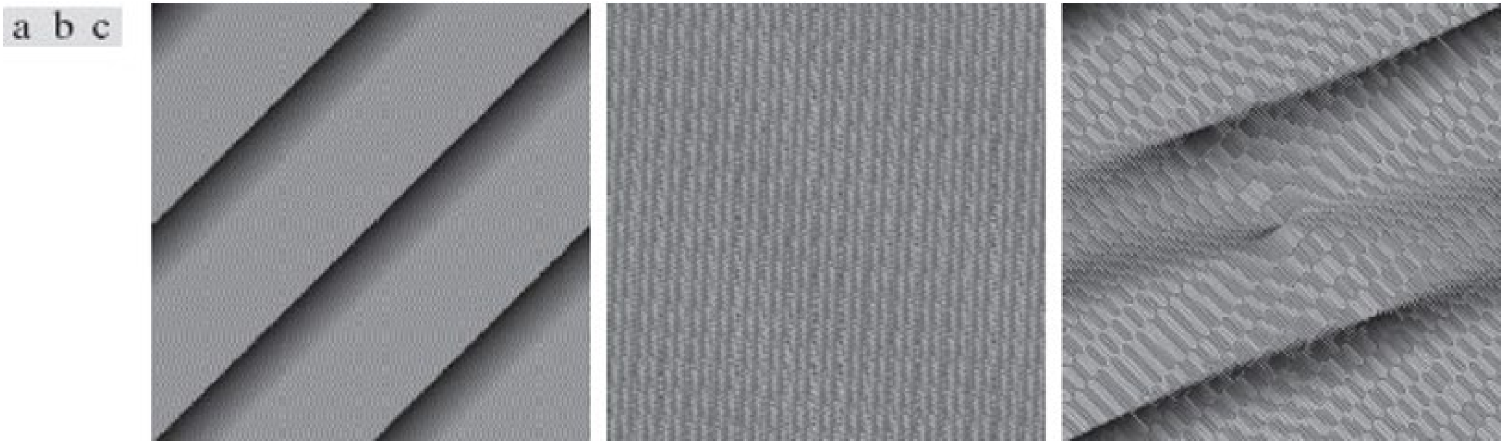


Figure 4.25

Phase angle images of (a) centered, (b) translated, and (c) rotated rectangles.



2-D Discrete Convolution Theorem

- In discrete space, the functions are periodic; and therefore, there convolution is also periodic. One period of this period, and thus called **circular convolution** is obtained by:

$$f(x) \star h(x) = \sum_{m=0}^{M-1} f(m)h(x-m) \qquad x = 0, 1, 2, \ldots, M-1$$

- For 2-D Circular Convolution:

$$(f \star h)(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n)h(x-m, y-n)$$

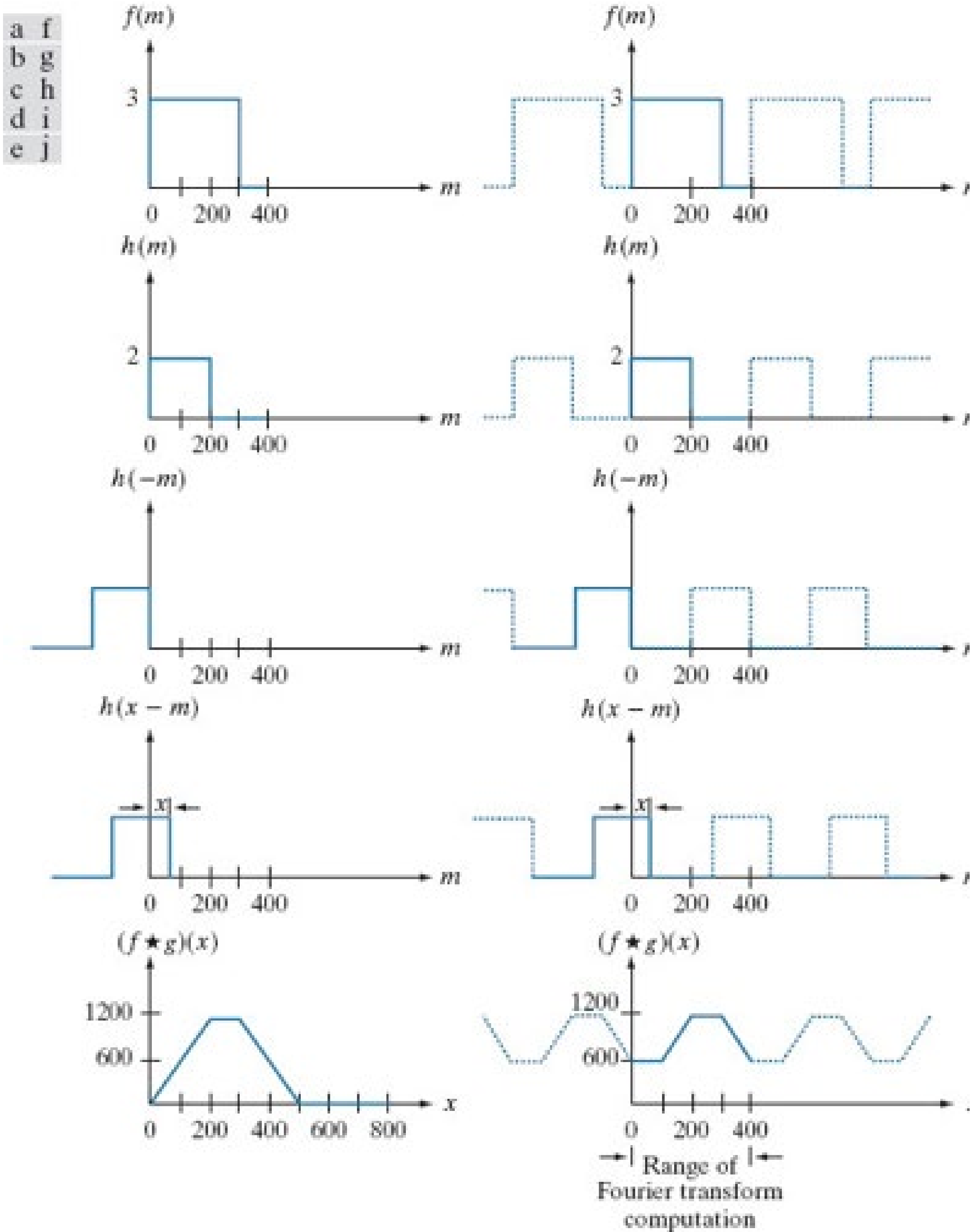
- We have:

$$(f \star h)(x, y) \Leftrightarrow (F \cdot H)(u, v)$$

$$(f \cdot h)(x, y) \Leftrightarrow \frac{1}{MN}(F \star H)(u, v)$$

Figure 4.27

Figure 4.27 Left column: Spatial convolution computed with Eq. (3-44), using the approach discussed in Section 3.4. Right column: Circular convolution. The solid line in (j) is the result we would obtain using the DFT, or, equivalently, Eq. (4-48). This erroneous result can be remedied by using **zero padding**.



Wraparound error & Zero Padding

- Using DFT and the convolution theorem, the periodicity inherent in the expression for the DFT must be considered.
- Using DFT, because we are convolving two periodic functions, the convolution itself is periodic. The closeness of the periods causes interference and the **wraparound error**.
- **Zero Padding (in 1D)**: If we append zeros to both functions so that they have the same length, denoted by P, then wraparound is avoided by choosing

$$P \geq A + B - 1$$

where f(x) and h(x) are composed of A and B samples respectively.

2-D Zero Padding

$$f_p(x,y) = \begin{cases} f(x,y) & 0 \le x \le A-1 \text{ and } 0 \le y \le B-1 \\ 0 & A \le x \le P \text{ or } B \le y \le Q \end{cases}$$

$$h_p(x,y) = \begin{cases} h(x,y) & 0 \le x \le C-1 \text{ and } 0 \le y \le D-1 \\ 0 & C \le x \le P \text{ or } D \le y \le Q \end{cases}$$

$$P \ge A + C - 1$$

$$Q \ge B + D - 1$$

- If two images of size M×N, smallest even padding:
  - P = 2M
  - Q = 2N

Table 4.3

Summary of DFT definitions and corresponding expressions.

Name	Expression(s)
1) Discrete Fourier transform (DFT) of $f(x,y)$	$F(u,v)=\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(ux/M+vy/N)}$
2) Inverse discrete Fourier transform (IDFT) of $F(u,v)$	$f(x,y)=\frac{1}{MN}\sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F(u,v)e^{j2\pi(ux/M+vy/N)}$
3) Spectrum	$\lvert F(u,v)\rvert = \left[R^2(u,v)+I^2(u,v)\right]^{1/2}$ $\quad R=\text{Real}(F);\ I=\text{Imag}(F)$
4) Phase angle	$\phi(u,v)=\tan^{-1}\left[\frac{I(u,v)}{R(u,v)}\right]$
5) Polar representation	$F(u,v)=\lvert F(u,v)\rvert e^{j\phi(u,v)}$
6) Power spectrum	$P(u,v)=\lvert F(u,v)\rvert^2$
7) Average value	$\bar{f}=\frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)=\frac{1}{MN}F(0,0)$
8) Periodicity (k_1 and k_2 are integers)	$F(u,v)=F(u+k_1M,v)=F(u,v+k_2N)$ $=F(u+k_1,v+k_2N)$ $f(x,y)=f(x+k_1M,y)=f(x,y+k_2N)$ $=f(x+k_1M,y+k_2N)$
9) Convolution	$(f\star h)(x,y)=\sum_{m=0}^{M-1}\sum_{n=0}^{N-1} f(m,n)h(x-m,y-n)$
10) Correlation	$(f \,☆\, h)(x,y)=\sum_{m=0}^{M-1}\sum_{n=0}^{N-1} f^*(m,n)h(x+m,y+n)$
11) Separability	The 2-D DFT can be computed by computing 1-D DFT transforms along the rows (columns) of the image, followed by 1-D transforms along the columns (rows) of the result. See Section 4.11.
12) Obtaining the IDFT using a DFT algorithm	$MNf^*(x,y)=\sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F^*(u,v)e^{-j2\pi(ux/M+vy/N)}$ This equation indicates that inputting $F^*(u,v)$ into an algorithm that computes the forward transform (right side of above equation) yields $MNf^*(x,y)$. Taking the complex conjugate and dividing by MN gives the desired inverse. See Section 4.11.

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Table 4.4

Summary of DFT pairs. The closed-form expressions in 12 and 13 are valid only for continuous variables. They can be used with discrete variables by sampling the continuous expressions.

Name	DFT Pairs
1) Symmetry properties	See Table 4.1
2) Linearity	$af_1(x,y)+bf_2(x,y) \Leftrightarrow aF_1(u,v)+bF_2(u,v)$
3) Translation (general)	$f(x,y)e^{j2\pi(u_0x/M+v_0y/N)} \Leftrightarrow F(u-u_0,v-v_0)$ $f(x-x_0,y-y_0) \Leftrightarrow F(u,v)e^{-j2\pi(ux_0/M+vy_0/N)}$
4) Translation to center of the frequency rectangle, $(M/2,N/2)$	$f(x,y)(-1)^{x+y} \Leftrightarrow F(u-M/2,v-N/2)$ $f(x-M/2,y-N/2) \Leftrightarrow F(u,v)(-1)^{u+v}$
5) Rotation	$f(r,\theta+\theta_0) \Leftrightarrow F(\omega,\varphi+\theta_0)$ $r=\sqrt{x^2+y^2} \quad \theta=\tan^{-1}(y/x) \quad \omega=\sqrt{u^2+v^2} \quad \varphi=\tan^{-1}(v/u)$
6) Convolution theorem†	$(f\star h)(x,y) \Leftrightarrow (F\bullet H)(u,v)$ $(f\bullet h)(x,y) \Leftrightarrow (1/MN)[(F\star H)(u,v)]$
7) Correlation theorem†	$(f\,☆\,h)(x,y) \Leftrightarrow (F^*\bullet H)(u,v)$ $(f^*\bullet h)(x,y) \Leftrightarrow (1/MN)[(F\,☆\,H)(u,v)]$
8) Discrete unit impulse	$\delta(x,y) \Leftrightarrow 1$ $1 \Leftrightarrow MN\delta(u,v)$
9) Rectangle	$\mathrm{rec}[a,b] \Leftrightarrow ab\dfrac{\sin(\pi ua)}{(\pi ua)}\dfrac{\sin(\pi vb)}{(\pi vb)}e^{-j\pi(ua+vb)}$
10) Sine	$\sin(2\pi u_0x/M+2\pi v_0y/N) \Leftrightarrow \dfrac{jMN}{2}[\delta(u+u_0,v+v_0)-\delta(u-u_0,v-v_0)]$
11) Cosine	$\cos(2\pi u_0x/M+2\pi v_0y/N) \Leftrightarrow \dfrac{1}{2}[\delta(u+u_0,v+v_0)+\delta(u-u_0,v-v_0)]$

The following Fourier transform pairs are derivable only for continuous variables, denoted as before by t and z for spatial variables and by μ and ν for frequency variables. These results can be used for DFT work by sampling the continuous forms.

12) Differentiation (the expressions on the right assume that $f(\pm\infty,\pm\infty)=0$.	$\left(\dfrac{\partial}{\partial t}\right)^m\left(\dfrac{\partial}{\partial z}\right)^n f(t,z) \Leftrightarrow (j2\pi\mu)^m(j2\pi\nu)^n F(\mu,\nu)$ $\dfrac{\partial^m f(t,z)}{\partial t^m} \Leftrightarrow (j2\pi\mu)^m F(\mu,\nu); \quad \dfrac{\partial^n f(t,z)}{\partial z^m} \Leftrightarrow (j2\pi\nu)^n F(\mu,\nu)$
13) Gaussian	$A2\pi\sigma^2e^{-2\pi^2\sigma^2(t^2+z^2)} \Leftrightarrow Ae^{-(\mu^2+\nu^2)/2\sigma^2}$ (A is a constant)

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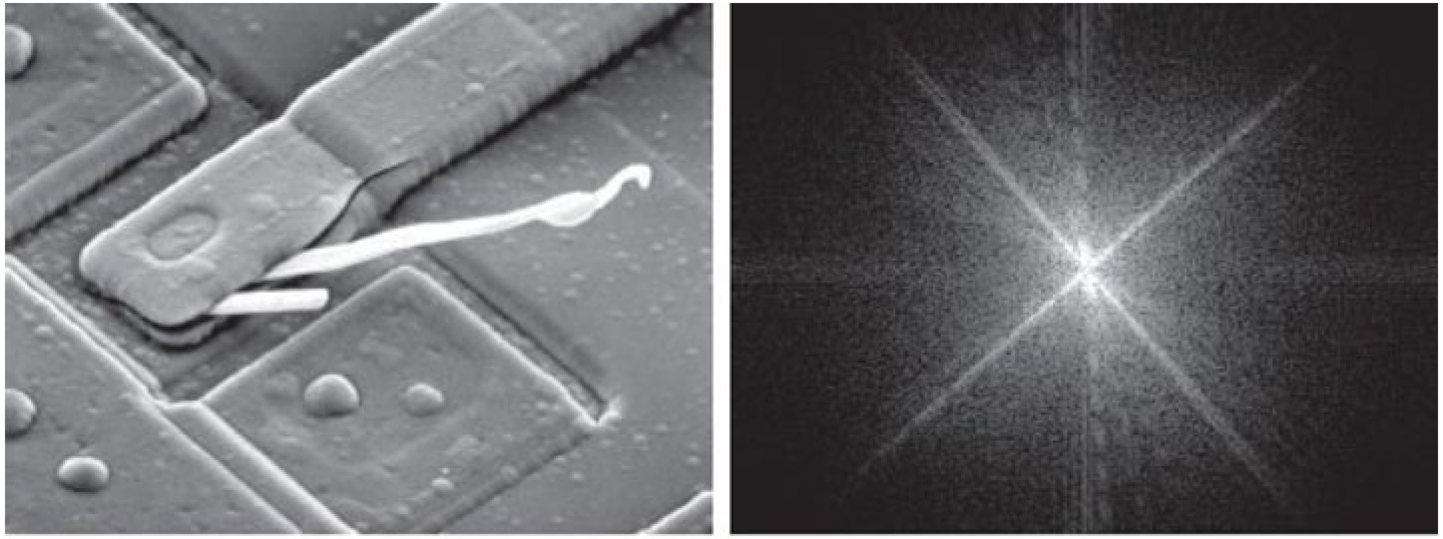
Filtering in the Frequency Domain

Figure 4.28

(a) SEM image of a damaged integrated circuit. (b) Fourier spectrum of (a). (Original image courtesy of Dr. J. M. Hudak, Brockhouse Institute for Materials Research, McMaster University, Hamilton, Ontario, Canada.)



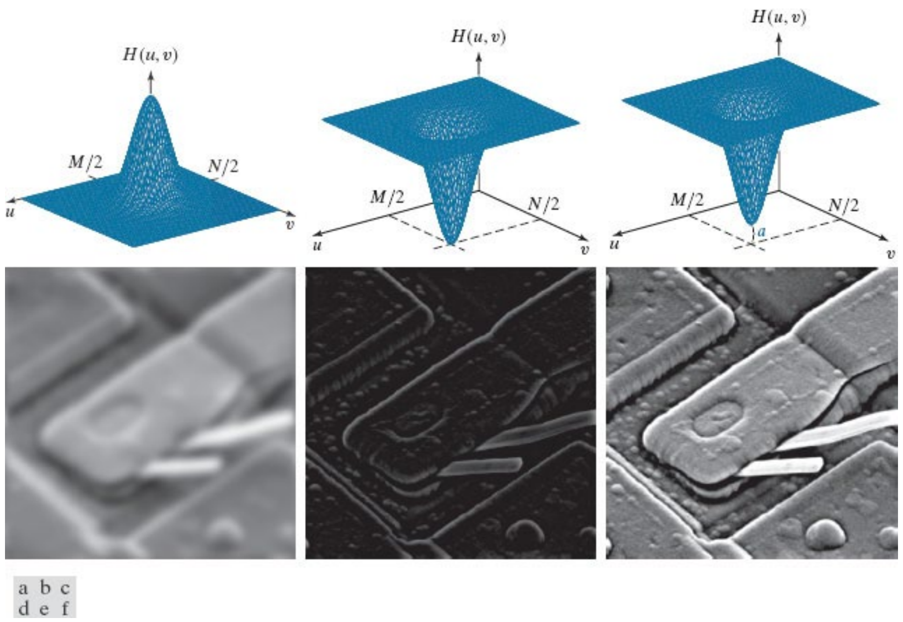
a b

Figure 4.29

Result of filtering the image in Fig. 4.28(a) with a filter transfer function that sets to 0 the dc term, $F(P/2, Q/2)$, in the centered Fourier transform, while leaving all other transform terms unchanged

Figure 4.30

Top row: Frequency domain filter transfer functions of (a) a lowpass filter, (b) a highpass filter, and (c) an offset highpass filter. Bottom row: Corresponding filtered images obtained using Eq. (4-104). The offset in (c) is **a** = 0.85, and the height of **H(u,v)** is 1. Compare (f) with Fig. 4.28(a).



Steps for Filtering in Frequency Domain

- To filter an image f(x,y) of size M × N:
  1. Obtain padding sizes, i.e., P = 2M and Q=2N
  2. Use zero, mirror, or replicate padding to obtain padded image $f_p(x,y)$ of size P × Q.
  3. To center the Fourier transform, multiply the padded image by $(-1)^{x+y}$.
  4. Compute DFT of the result of step 3, F(u,v).
  5. Construct a real, symmetric filter transfer function, H(u,v) of size H(u,v), centered at (P/2,Q/2).
  6. Calculate the elementwise multiplication G(u,v) = H(u,v).F(u,v).
  7. Calculate the real component of IDFT of G(u,v), and undo step 3:
  $$g_p(x,y)=\left(\text{real}\left[\Im^{-1}\{G(u,v)\}\right]\right)(-1)^{x+y}$$
  8. Extract the top left M × N region to obtain the final filtered result, g(x,y).

Figure 4.35

(a) An $M \times N$ image, f. (b) Padded image, f_p of size $P \times Q$. (c) Result of multiplying f_p by $(-1)^{x+y}$. (d) Spectrum of **F**. (e) Centered Gaussian lowpass filter transfer function, **H**, of size $P \times Q$. (f) Spectrum of the product **HF**. (g) Image the real part of the IDFT of **HF**, multiplied by $g_p(-1)^{x+y}$. (h) Final result, **g**, obtained by extracting the first **M** rows and **N** columns of g_p.

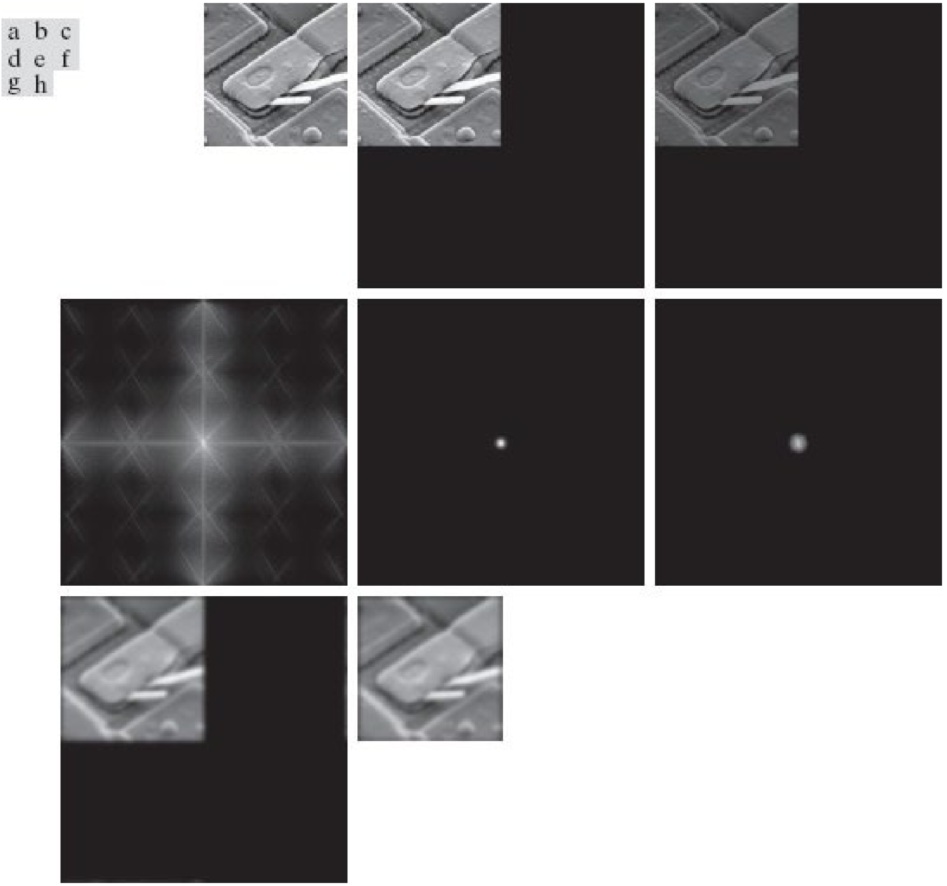


Figure 4.36

lowpass transfer function in the frequency domain. (b) Corresponding kernel in the spatial domain. (c) Gaussian highpass transfer function in the frequency domain. (d) Corresponding kernel. The small 2-D kernels shown are kernels we used in Chapter 3.

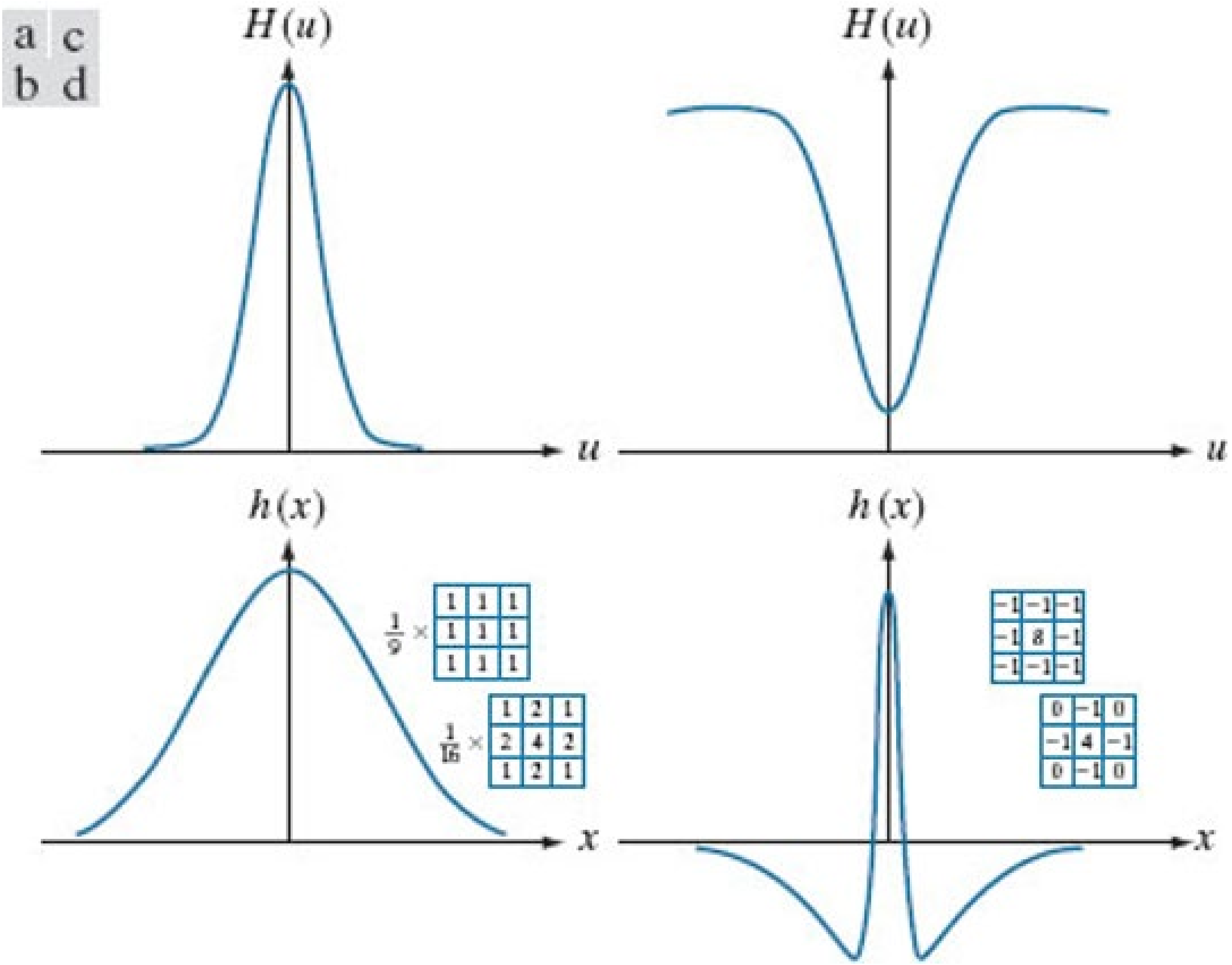


Figure 4.37

(a) Image of a building, and (b) its Fourier spectrum.

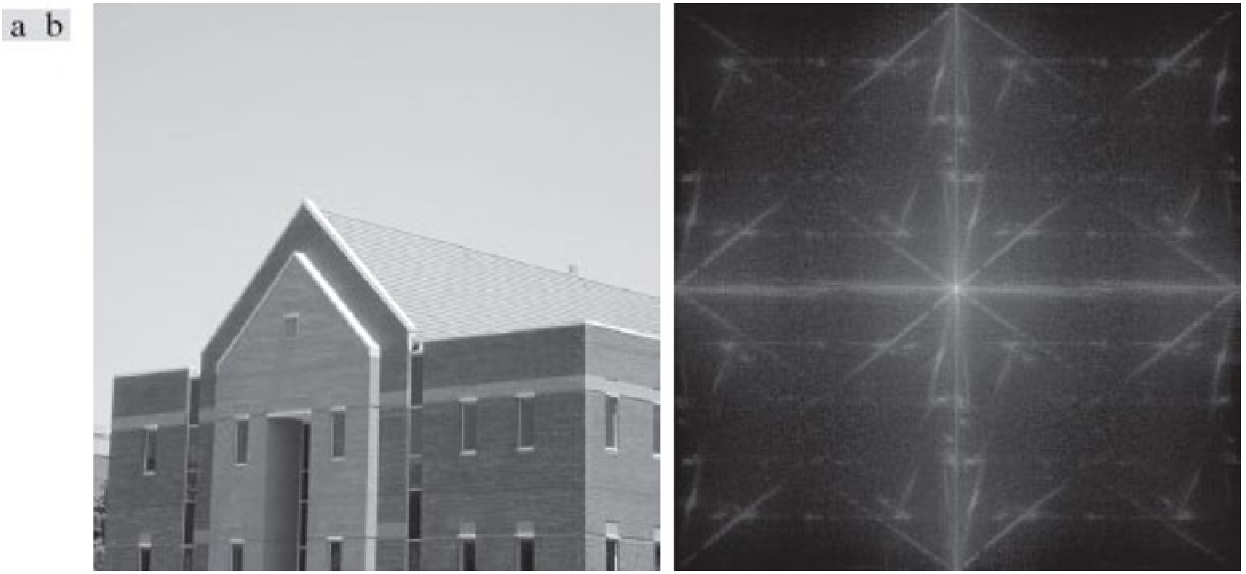


Figure 4.38

(a) A spatial kernel and perspective plot of its corresponding frequency domain filter transfer function. (b) Transfer function shown as an image. (c) Result of filtering Fig. 4.37(a) in the frequency domain with the transfer function in (b). (d) Result of filtering the same image in the spatial domain with the kernel in (a). The results are identical.

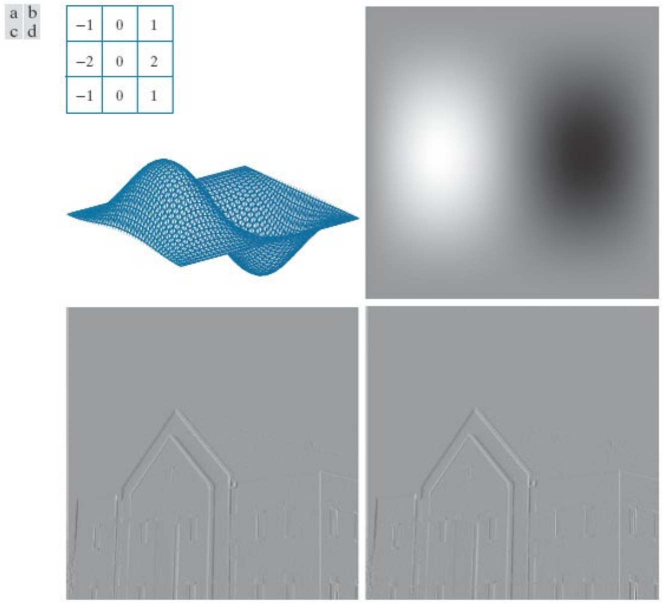
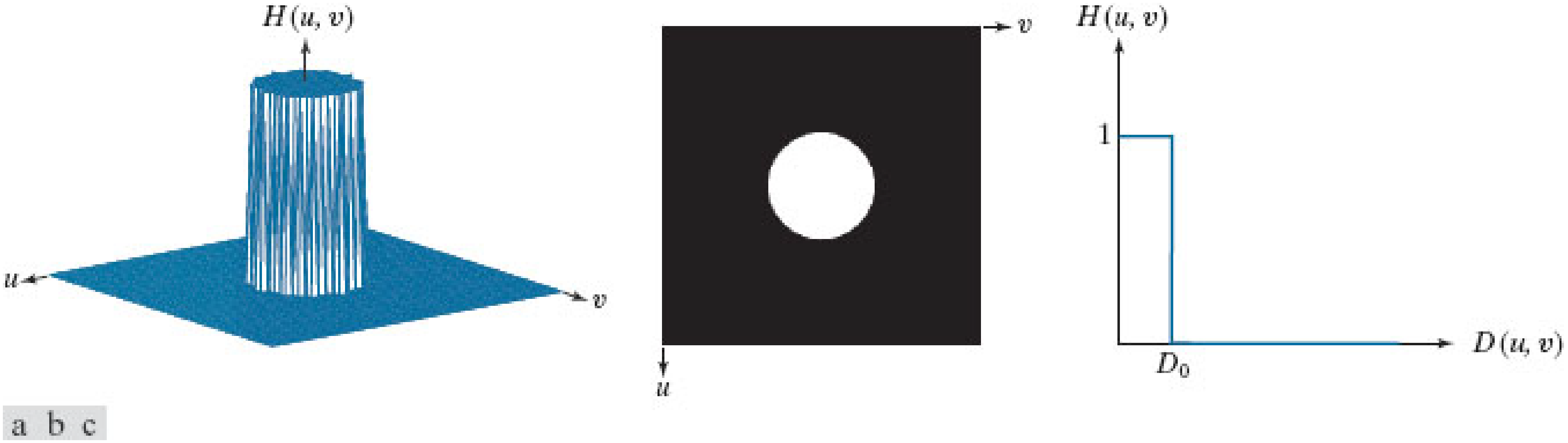


Figure 4.39

(a) Perspective plot of an ideal lowpass-filter (ILPF) transfer function. (b) Function displayed as an image. (c) Radial cross section.

$$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \le D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$$



Pearson

Figure 4.42

(a) Frequency domain ILPF transfer function. (b) Corresponding spatial domain kernel function. (c) Intensity profile of a horizontal line through the center of (b).

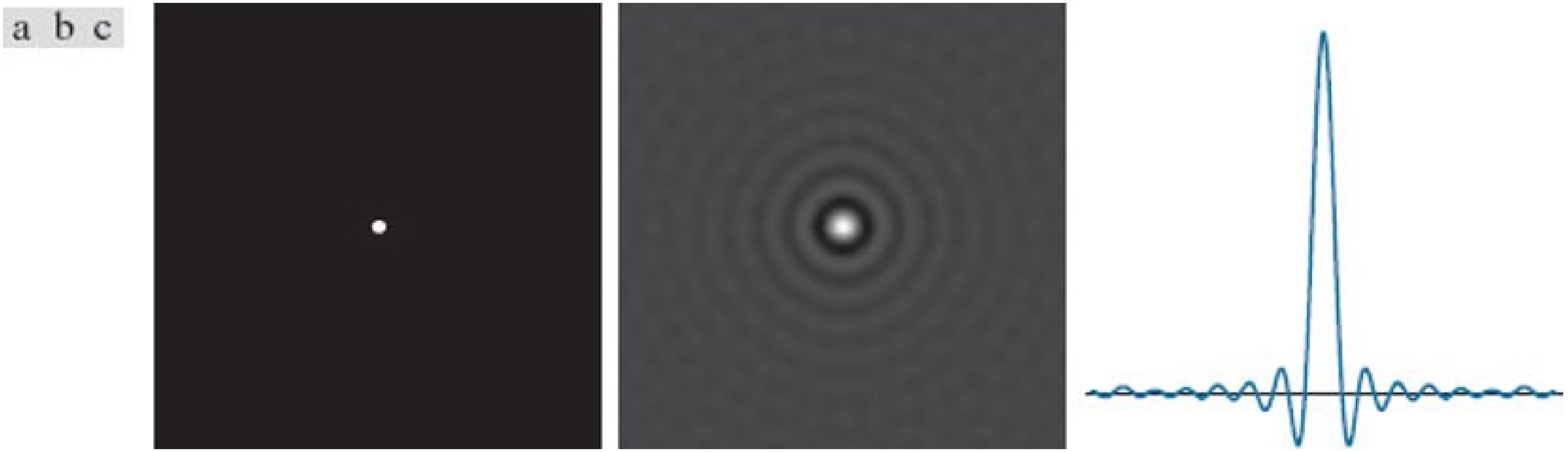
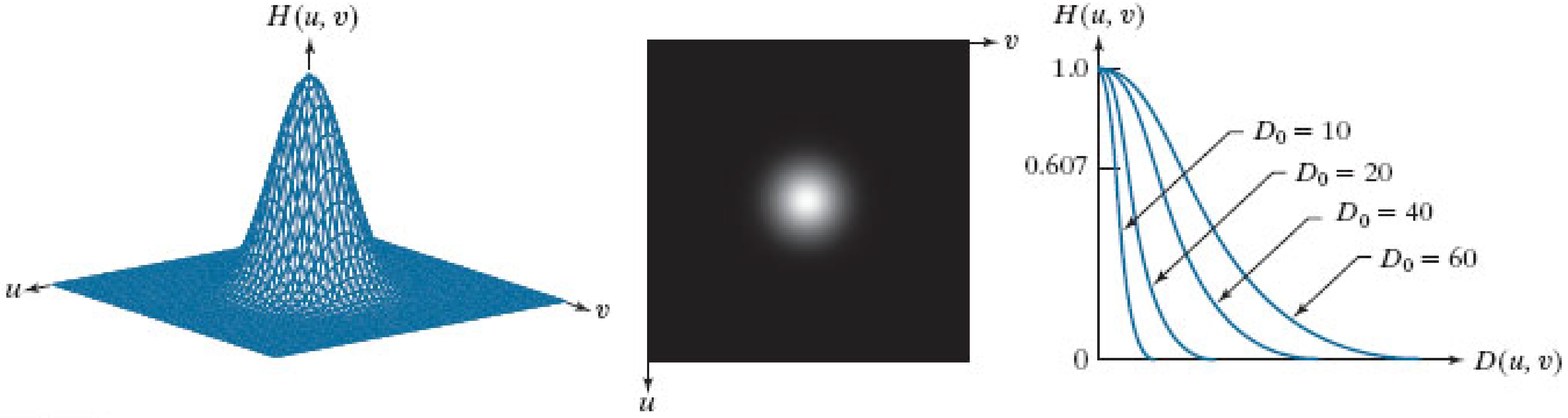


Figure 4.43

(a) Perspective plot of a Gaussian lowpass filter (GLPF) transfer function. (b) Function displayed as an image. (c) Radial cross sections for various values of D_0.

$$H(u,v) = e^{-D^2(u,v)/2\sigma^2}$$



a b c

Figure 4.45

(a) Perspective plot of a Butterworth lowpass-filter (BLPF) transfer function. (b) Function displayed as an image. (c) Radial cross sections of BLPFs of orders 1 through 4.

$$H(u,v) = \frac{1}{1 + [D(u,v)/D_0]^{2n}}$$

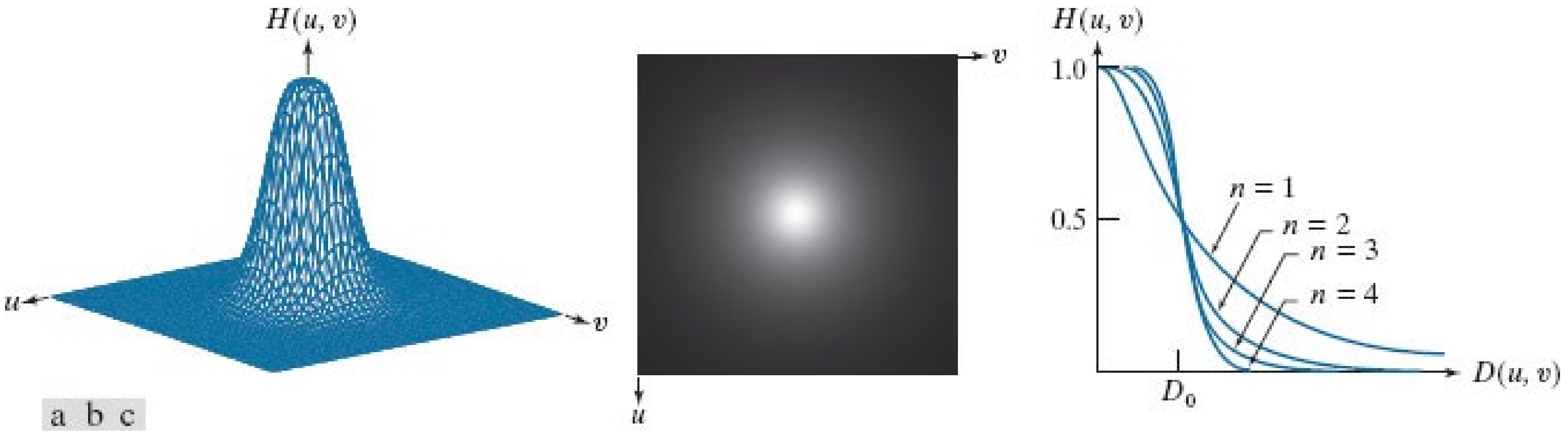


Table 4.5

Lowpass filter transfer functions. D_0 is the cutoff frequency, and n is the order of the Butterworth filter.

Ideal	Gaussian	Butterworth
$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \le D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$	$H(u,v) = e^{-D^2(u,v)/2D_0^2}$	$H(u,v) = \dfrac{1}{1 + [D(u,v)/D_0]^{2n}}$

Figure 4.49

(a) Original 785×732 image. (b) Result of filtering using a GLPF with $D_0 = 150$. (c) Result of filtering using a GLPF with $D_0 = 130$. Note the reduction in fine skin lines in the magnified sections in (b) and (c).

Figure 4.51 – High Pass Filters

Top row: Perspective plot, image, and, radial cross section of an IHPF transfer function. Middle and bottom rows: The same sequence for GHPF and BHPF transfer functions. (The thin image borders were added for clarity. They are not part of the data.)

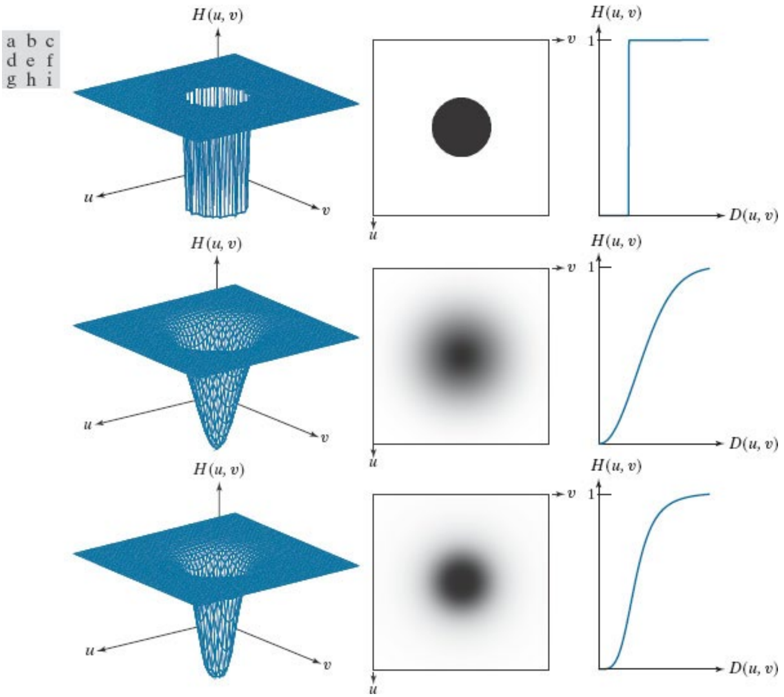


Figure 4.52

(a)–(c): Ideal, Gaussian, and Butterworth highpass spatial filters obtained from IHPF, GHPF, and BHPF frequency-domain transfer functions. (The thin image borders are not part of the data.) (d)–(f): Horizontal intensity profiles through the centers of the kernels.

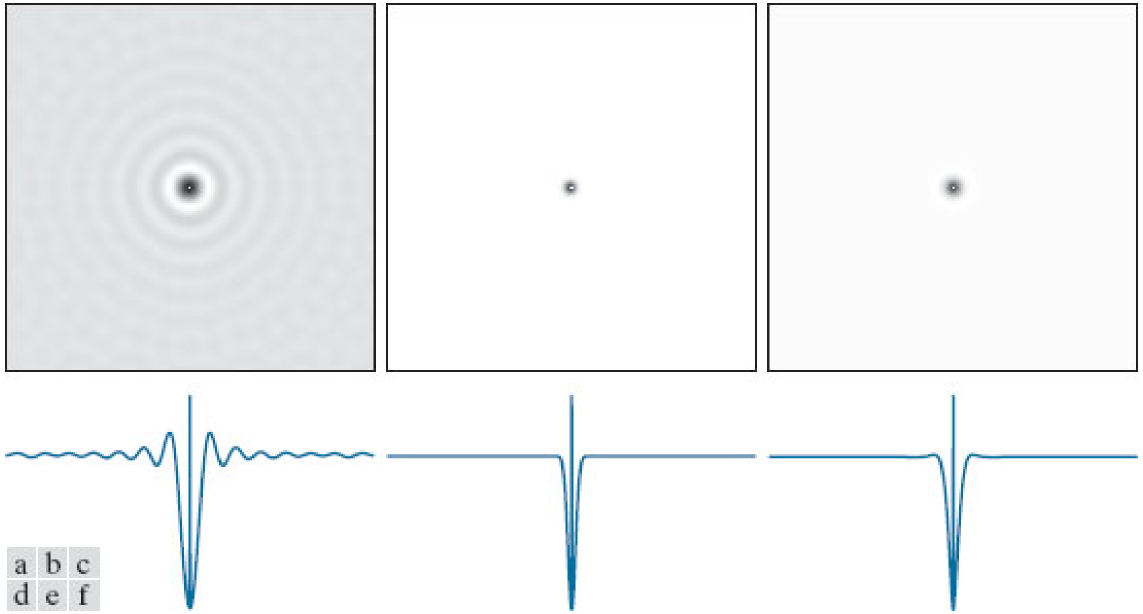


Table 4.6

Highpass filter transfer functions. D_0 is the cutoff frequency and n is the order of the Butterworth transfer function.

Ideal	Gaussian	Butterworth
$H(u,v) = \begin{cases} 0 & \text{if } D(u,v) \le D_0 \\ 1 & \text{if } D(u,v) > D_0 \end{cases}$	$H(u,v) = 1 - e^{-D^2(u,v)/2D_0^2}$	$H(u,v) = \dfrac{1}{1 + [D_0/D(u,v)]^{2n}}$

Pearson

Figure 4.55

(a) Smudged thumbprint. (b) Result of highpass filtering (a). (c) Result of thresholding (b). (Original image courtesy of the U.S. National Institute of Standards and Technology.)

a b c

Figure 4.56

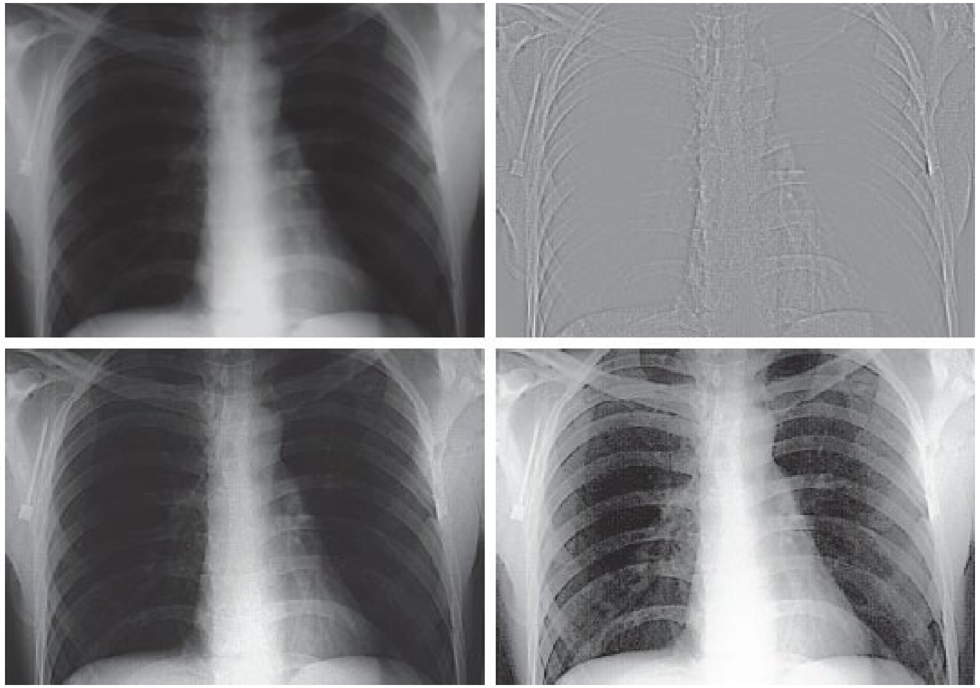
(a) Original, blurry image. (b) Image enhanced using the Laplacian in the frequency domain. Compare with Fig. 3.52(d). (Original image courtesy of NASA.)

a b

Figure 4.57

(a) A chest X-ray. (b) Result of filtering with a GHPF function. (c) Result of high-frequencyemphasis filtering using the same GHPF. (d) Result of performing histogram equalization on (c). (Original image courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School.)

a b
c d



Summary

- The 2-D sampling theorem states that a continuous band-limited function f(t,z) can be recovered with no error from its samples, if the sampling rates in each dimension exceeds twice the highest frequency content in that dimension (Nyquist rate).
- To apply filtering in the frequency domain, zero-padding and centering must be applied first. Instead of convolving in the spatial domain, we can multiply the Fourier transforms of the kernel and the image.
- Three types of low pass filters are Ideal (ILPF), Gaussian (GLPF), and Butterworth (BLPF) filters. Similarly, three types of high pass filters are Ideal (IHPF), Gaussian (GHPF), and Butterworth (BHPF) filters.

References

- [DIP] Gonzalez, R.C. and Woods, R.E., Digital Image Processing, Pearson, 4th edt., 2018.
- [CV] Computer Vision: Algorithms and Applications, R. Szeliski (http://szeliski.org/Book)
- [PICV] Practical introduction to Computer Vision with OpenCV, Kenneth Dawson-Howe, WILEY, 2014. Available through Seneca libraries

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